Prioritizing potential aquaculture for Oyster and California Mussel

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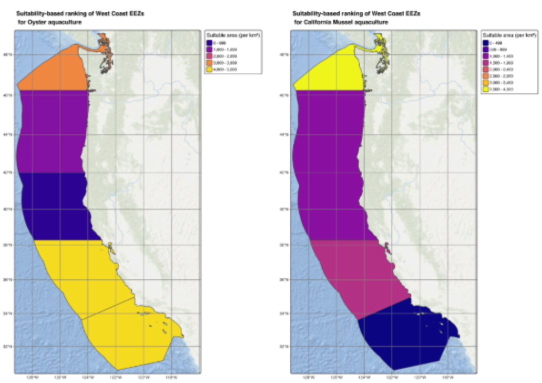
Screenshot of GitHub README.md

Prioritizing potential aquaculture for Oyster and California Mussel

This repository contains the details of the assignment 4 for the course EDS 223: Geospatial Analysis & Remote Sensing for the Master of Environmental Data Science (MEDS) program.

The assignment focuses on determining which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited to developing marine aquaculture for several species of Oysters and a California Mussel.



Background

Marine aquaculture has the potential to play an important role in the global food supply as a more sustainable protein option than land-based meat production. Gentry et al. mapped the potential for marine aquaculture globally based on multiple constraints, including ship traffic, dissolved oxygen, and bottom depth. They found that global seafood demand could be met using less than 0.015% of the global ocean area2.

All analysis were done on R version 4.5.1 using the following libraries tidyverse, sf, tmap, terra and kableExtra.

Objective of the assignment

Determine the Exclusive Economic Zones (EEZ) on the West Coast of the US best suited to develop marine aquaculture for several species of oysters and a species of our choice. For this, - create a function that has the following characteristics: - arguments: minimum and maximum sea surface temperature minimum and maximum depth species name - outputs: map of EEZ regions colored by amount of suitable area species name should be included in the map's title

Contents

This repository contains the following files 1. .gitignore 2. prioritizing-potential-aquaculture.Rproj 3. aquaculture_analysis.qmd 4. figs: visual representation that is derived from the raw data 5. README.md

File Structure

```
EDS223-HW3
└── README.md
└── prioritizing-potential-aquaculture.Rproj
└── aquaculture_analysis.qmd
└── aquaculture_analysis.pdf
└── figs
    └── california_mussel_map.png
    └── combined_aquaculture_map.png
    └── oyster_eez_map.png
└── .gitignore
    └──data
        └── wc_regions_clean.shp
        └── depth.tif
        └── average_annual_sst_2008.tif
        └── average_annual_sst_2009.tif
        └── average_annual_sst_2010.tif
        └── average_annual_sst_2011.tif
        └── average_annual_sst_2012.tif
```

Data

Due to its large size, the `data` folder is included in `.gitignore` and is not tracked by version control. For this assignment, the data were pre-downloaded and provided by the team. Details of the data sources and download links are as follows:

Data download link

To access the data, click here

Data Source

Species You can find information on species depth and temperature requirements on SeaLifeBase. We are thinking about the potential for marine aquaculture, so these species should have some reasonable potential for commercial consumption.

Sea Surface Temperature We are using average annual sea surface temperature (SST) from the years 2008 to 2012 to characterize the average sea surface temperature within the region. The data we are working with was originally generated from NOAA's 5km Daily Global Satellite Sea Surface Temperature Anomaly v3.1.

Bathymetry To characterize the depth of the ocean we are using the General Bathymetric Chart of the Oceans (GEBCO).

Exclusive Economic Zones We designate maritime boundaries using Exclusive Economic Zones off of the west coast of US from Marineregions.org.

Course Information

- **Course Title:** EDS 223 - Geospatial Analysis & Remote Sensing
- **Term:** Fall 2025
- **Program:** UCSB Masters in Environmental Data Science.

Teaching Team:

- **Instructor:** Annie Adams
- **Teaching Assistant:** Alessandra Vidal Meza

Complete description for the homework can be found on the Assignment-4

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Load required libraries

```r
library(tidyverse)
library(sf)
library(tmap)
library(terra)
library(kableExtra)
```

Read data

```r
# Read sea surface temperature data
sst_2008 <- rast(here::here('data', 'average_annual_sst_2008.tif'))
sst_2009 <- rast(here::here('data', 'average_annual_sst_2009.tif'))
sst_2010 <- rast(here::here('data', 'average_annual_sst_2010.tif'))
sst_2011 <- rast(here::here('data', 'average_annual_sst_2011.tif'))
sst_2012 <- rast(here::here('data', 'average_annual_sst_2012.tif'))

# Combine SST raster into a raster stack
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012,
               along = list(year = 2008:2012))

# Read Bathymetry data
depth <- rast(here::here('data', 'depth.tif'))

# Read Exclusive Economic Zones data
eez <- st_read(here::here('data', 'wc_regions_clean.shp'))
```

Prepare data

```r
# Check CRS for all data set
st_crs(sst_stack)
st_crs(depth)
st_crs(eez)
```

```r
# Convert the CRS of sst_stack to EPSG:4326
crs(sst_stack) <- "EPSG:4326"
```

```
# Check CRS of the dataset
if(st_crs(sst_stack) == st_crs(depth)){
print("Coordinate reference systems match!")
} else {
warning("Update coordinate reference systems to match!")
}
```

```
[1] "Coordinate reference systems match!"
```

Process data

Find the mean of SST stacks from 2008-2012

```
# Find the mean SST from 2008-2012 (e.g. create single raster of average SST)
sst_mean <- app(sst_stack, mean, na.rm = TRUE)

# Convert average SST from Kelvin to Celsius
sst_mean_celsius <- sst_mean - 273.15
```

Crop depth raster to match the extent of the SST raster

```
# Crop the ocean depth
depth_cropped <- crop(depth, ext(sst_mean_celsius))
```

Resample the depth data to match the resolution of the SST data using the nearest neighbor approach

```
# Resample the ocean depth data
depth_resample <- resample(depth_cropped, sst_mean_celsius, method = 'near')
```

Check that the ocean depth and SST match in resolution, extent and coordinate reference system - *Can the rasters be stacked?*

```
# Can the rasters be stacked?
if(all.equal(ext(depth_resample), ext(sst_mean_celsius)) == TRUE &&
   all.equal(res(depth_resample), res(sst_mean_celsius)) == TRUE &&
   crs(depth_resample) == crs(sst_mean_celsius)) {
  test_stack <- c(depth_resample, sst_mean_celsius)
  print('Rasters can be successfully stacked!')
} else {
  warning('Rasters cannot be stacked. Compatibility issue detected!')
}
```

```
[1] "Rasters can be successfully stacked!"
```

Find suitable locations for Oysters

Reclassify SST and ocean depth data into locations that are suitable for Oysters

```r
# Reclassify SST into locations that are suitable for oysters
sst_reclass_oysters <- matrix(c(-Inf, 11, 0,
                         11, 30, 1,
                         30, Inf, 0), ncol = 3, byrow = TRUE)

sst_classify_oysters <- classify(sst_mean_celsius, sst_reclass_oysters)

depth_reclass_oysters <- matrix(c(-Inf, -70, 0,
                         -70, 0, 1,
                         0, Inf, 0), ncol = 3, byrow = TRUE)

depth_classify_oysters <- classify(depth_resample, depth_reclass_oysters)

# Find suitable locations
suitable_locations_oysters <- sst_classify_oysters * depth_classify_oysters
```

Determine the most suitable EEZ

Select suitable cells within West Coast EEZs

```r
# Check and transform coordinate reference systems
if(st_crs(eez) == crs(sst_mean_celsius)) {
  print("Coordinate reference systems match!")
} else {
  warning("Updating coordinate reference systems to match!")
}
```

```
Warning: Updating coordinate reference systems to match!
```

```r
# Transform economic zones data set to match raster CRS
eez <- st_transform(eez, crs(sst_mean_celsius))
```

```r
# Convert the eez polygons to raster using the regions
eez_raster <- rasterize(eez, suitable_locations_oysters, field = "rgn_id")
```

Find area of grid cells

```r
# Calculate the area in km square
area_km2_cell <- cellSize(suitable_locations_oysters, unit = "km")
```

Find the total suitable area within each EEZ

```r
# Calculate the suitable area
suitable_area <- suitable_locations_oysters * area_km2_cell

# Calculate suitable area for each eez polygons
eez_suitable_area <- extract(suitable_area, eez, fun = "sum", na.rm = TRUE)

# Add the eez suitable area to the eez dataset
eez$suitable_area_km2 <- eez_suitable_area[, 2]

# Sort by suitable area (descending)
eez_rank <- eez[order(-eez$suitable_area_km2), ]
```

```r
# Convert to data frame and select columns
eez_df <- as.data.frame(eez_rank)[, c("rgn", "suitable_area_km2")]

# Visualize the eez rank in table format
kable(eez_df,
      caption = "Suitable economic zones in West Coast for Oyster aquaculture",
      col.names = c("Region", "Suitable area (per km²)"),
      digits = 2,
      row.names = FALSE,
      format.args = list(big.mark = ",")) %>%
  kable_styling(full_width = FALSE, font_size = 11)
```

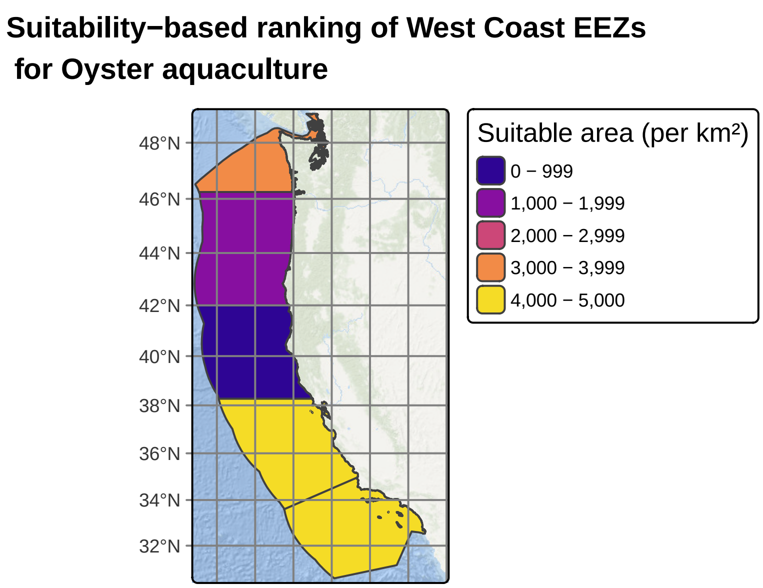
Table 1: Suitable economic zones in West Coast for Oyster aquaculture

Region	Suitable area (per km^2)
Central California	4,923.15
Southern California	4,096.54
Washington	3,224.74

Oregon	1,533.09
Northern California	438.15

```
# Plot the suitability ranking for Oyster aquaculture
oyster_map <- tm_shape(eez_rank) +
  tm_polygons(fill = "suitable_area_km2",
              fill.scale = tm_scale(values = "plasma"),
              fill.legend = tm_legend(title = "Suitable area (per km²)",
                                      fontface = "bold",
                                      title.size = 0.8,
                                      text.size = 0.6)) +
  tm_title(text = "Suitability-based ranking of West Coast EEZs\n for Oyster aquaculture",
           size = 0.9, fontface = "bold") +
  tm_graticules() +
  tm_basemap(server = "Esri.OceanBasemap", alpha = 0.8) +
  tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
            component.autoscale = FALSE)

oyster_map
```



Building a generalizable workflow

Here, we'll combine the above workflow above into a single function to ensure the process is clear and reproducible.

I will be determining which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited for developing marine aquaculture for **California mussel** *(Mytilus californianus)*. The suitable sea surface temperature (SST) is usually 8.6 - 14.3 degree Celsius and depth values is 0 - 30 meters for this species.

```
# Function to find suitable aquaculture locations and create outputs
aquaculture_location <- function(species_name, min_temp, max_temp,
                                 min_depth, max_depth)
  {
  # Reclassify SST based on temperature range
  sst_reclass <- matrix(c(-Inf, min_temp, 0,
                          min_temp, max_temp, 1,
                          max_temp, Inf, 0),
                        ncol = 3, byrow = TRUE)
  sst_classify <- classify(sst_mean_celsius, sst_reclass)

  # Reclassify depth based on depth range
  depth_reclass <- matrix(c(-Inf, min_depth, 0,
                            min_depth, max_depth, 1,
                            max_depth, Inf, 0),
                          ncol = 3, byrow = TRUE)
  depth_classify <- classify(depth_resample, depth_reclass)

  # Find suitable locations
  suitable_locations <- sst_classify * depth_classify

  # Convert EEZ polygons to raster
  eez_raster <- rasterize(eez, suitable_locations, field = "rgn_id")

  # Calculate area in km²
  area_km2_cell <- cellSize(suitable_locations, unit = "km")

  # Calculate suitable area
  suitable_area <- suitable_locations * area_km2_cell

  # Calculate suitable area for each EEZ polygon
  eez_suitable_area <- extract(suitable_area, eez, fun = "sum", na.rm = TRUE)
```

```r
  # Add suitable area to EEZ dataset
  eez$suitable_area_km2 <- eez_suitable_area[, 2]

  # Sort by suitable area (descending)
  eez_rank <- eez[order(-eez$suitable_area_km2), ]

  # Convert to data frame and select columns
  eez_df <- as.data.frame(eez_rank)[, c("rgn", "suitable_area_km2")]

  # Visualize the eez rank in table format
  eez_table <- kable(eez_df,
                     caption = paste0("Suitable economic zones in West Coast for ", species_n
                     col.names = c("Region", "Suitable area (per km²)"),
                     digits = 2,
                     row.names = FALSE,
                     format.args = list(big.mark = ",")) %>%
    kable_styling(full_width = FALSE, font_size = 11)

  # Create suitability map
  species_map <- tm_shape(eez_rank) +
    tm_polygons(fill = "suitable_area_km2",
                fill.scale = tm_scale(values = "plasma"),
                fill.legend = tm_legend(title = "Suitable area (per km²)",
                                        fontface = "bold",
                                        title.size = 0.8,
                                        text.size = 0.6)) +
    tm_title(text = paste0("Suitability-based ranking of West Coast EEZs\n for ",
                           species_name, " aquaculture"),
             size = 0.9, fontface = "bold") +
    tm_graticules() +
    tm_basemap(server = "Esri.OceanBasemap", alpha = 0.8) +
    tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
              component.autoscale = FALSE)

  # Return both table and map as a list
  return(list(table = eez_table, map = species_map))
}
```

Apply function for species of choice

```r
# Apply function for California Mussel
california_mussel <- aquaculture_location(species_name = "California Mussel",
  min_temp = 8.6,
  max_temp = 14.3,
  min_depth = -30,
  max_depth = 0)

# Display the suitability table for California Mussel
california_mussel$table
```

Table 2: Suitable economic zones in West Coast for California Mussel aquaculture

Region	Suitable area (per km^2)
Washington	3,603.10
Central California	1,843.34
Oregon	1,443.52
Northern California	1,155.22
Southern California	318.81

```r
# Display the suitability map for California Mussel
california_mussel$map
```

Suitability−based ranking of West Coast EEZs for California Mussel aquaculture